One selected state for higher Wick descendants

Boundary sources, charged towers and fixed-degree energy pinching

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Abstract

We extend the single-excitation boundary identity from the Hamiltonian, size fluctuation and quadratic curvature to disjoint marked Wick blocks of arbitrary fixed lengths, including neutral marked subblocks and whole-composite Gaussian-chaos projections. The proof reduces every source collision to at most four endpoints. A finite exhaustive rational certificate verifies its local algebra. The complete charged Wick tower has the same reduced density $1-\sqrt{q}+\sqrt{q}\,h^2$. Combining the all-unrestricted-source identity with existing charged spectral estimates transfers exact energy pinching to each fixed descendant in the selected state. The degree-uniform ordinary charged bridge remains a separate task.

1 Model and admissible expressions

Use standard CAR, the overlapping Hermitian complex Gaussian p-body Hamiltonian H, one common disorder, a fixed half-filled Slater vector O, and $m = N/2$. Set $\lambda_N = p^2/N \to \lambda > 0$, $C_Q = \mathbb{E}H^2$, and let C_m be its sector value. Define

$$DX = QX - \sum_i c_i^\dagger X c_i, \qquad \mathcal{L}X = \tfrac{1}{2}(NX - \sum_i (c_i^\dagger X c_i + c_i X c_i^\dagger)),$$
$$h_N = H_m/\sqrt{C_m}, \qquad W_N = C_m^{-1/2}(p - \mathcal{L} - \lambda_N)H|_m.$$

Loss maps act before restricting charge. The normalization C_m is a fixed target-sector scalar, not the operator C_Q inside a loss map. $M_N^\dagger$ is the previously normalized same-disorder occupied-to-empty block, with $\mathbb{E}\langle O|M_N M_N^\dagger|O\rangle = 1$. Time is dimensionless h-time and $\hbar = 1$.

In an ordered list of H slots, a mark (a, b, T) means the deficit $((b-a)p - T)$ on that entire interval, for $T = D, \mathcal{L}$. Marked intervals are disjoint and contain no complete Gaussian pair. Larger Wick regions can enforce stronger exclusions. A chaos region (a, b, d) keeps exactly $(b-a-d)/2$ internal pairs. These constraints implement, for example,

$$\mathcal{W}_d\big[H^a\big(((vp - \mathcal{L})Z_v) \diamond Z_w\big)\big], \qquad Z_v = \mathcal{W}_v(H^v).$$

Here $\diamond$ is coupling Wick multiplication. An arbitrary fixed scalar shift of each deficit and finite linear combinations are allowed. Repeated overlapping loss maps are not part of this expression class.

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2 General source and the four-endpoint reduction

For a pairing π with k edges, the right endpoint of edge e is $C_{P_e}^\dagger C_{Q_e}$ and the left is its adjoint. A G edge is unrestricted; an F edge has P_e empty and Q_e occupied relative to O. Attach square-zero commuting variables z_α to the marks. At a site, multiply the whole marked block before applying $1 - zT_i$. The local maps on a matrix are

$$D_i \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 0 & b/2 \\ c/2 & d-a \end{pmatrix}, \qquad \mathcal{L}_i \begin{pmatrix} a & b \\ c & d \end{pmatrix} = \frac{1}{2} \begin{pmatrix} a-d & b \\ c & d-a \end{pmatrix}.$$

Take the empty/occupied diagonal entry, keep the selected membership constraint, and multiply by the Jordan–Wigner crossing sign. Denote the result by $f_o(S; z)$, where S is the set of present P/Q labels. For one label i, its correct-occupancy singleton is $s_i(z)$, with $s_i(0) = 1$. Set $F_o(S; z) = f_o(S; z)/\prod_{i\in S} s_i(z)$. Then

$$K_\pi(z) = 2\sum_{i<j} \sum_{o=0,1} F_o(\{i, j\}; z) - 2k(k-1) - 2k_G. \tag{1}$$

With $\gamma = e^{-\lambda}$ and $q = \gamma^4$, a diagram contributes

$$\gamma^{b_{\mathrm{mix}}} [z_1 \cdots z_r] \exp\left(\lambda K_\pi(z) - \sum_\alpha \mu_\alpha z_\alpha\right), \qquad K_\pi(0) = -4\nu_\pi. \tag{2}$$

ν_π counts crossings and nestings with an inner F edge; b_{mix} counts mixed ordinary/selected edges.

The earlier fixed-Slater source derivation [3] extends without a length-two assumption. No pair inside a mark means every single present label gives only an odd local factor there. Thus $\prod_i s_i^p \exp[p\sum_\alpha (b_\alpha - a_\alpha) z_\alpha] = 1$ exactly. For each fixed layout, normalized defect coefficients have bounded nilpotent norm B. The old falling-factorial domination applies with activity $Bm(p/m)^{|S|}$: its tails are summable, triples disappear after domination, and backward selected edges require a defect of order p. Wick and chaos restrictions only delete pairings. This establishes (2) for every fixed admissible expression.

Lemma 1 (Finite local reduction)**.** *Every source-dependent summand of* (1) *is an own-edge collision with two endpoints or a two-edge collision with four endpoints. Deleting all other endpoints preserves its normalized local polynomial.*

Proof. Only the two present labels can supply nonidentity site matrices. An empty marked interval contributes $T_i(I) = 0$. Each surviving marked interval becomes a contiguous interval among the retained endpoints; disjointness and absence of an internal pair survive. The singleton denominator and crossing sign use exactly the same present labels. An F edge touching an extreme endpoint remains extreme after deletion. Thus the ordered local cases have no dependence on the original lengths. □

3 Local identities and the selected density

For an own-edge collision, let ℓ_1, ℓ_2 be the source variables at its endpoints when the corresponding mark is of type L, and zero otherwise. The source-dependent contribution is

$$K^G_{\text{own}} = \ell_1 + \ell_2 + \ell_1\ell_2, \qquad K^F_{\text{own}} = 0.$$

The endpoints cannot belong to the same mark. In a boundary-selected word $MBM^\dagger$, at least one endpoint of every F edge is outside all interior marks, so its own-edge difference is just $-\ell_1 - \ell_2$.

For distinct edges, replacing a G edge that touches a selected exterior endpoint by F leaves every nonconstant collision coefficient unchanged. Also, an all-G L singleton placed at an exterior endpoint has no source-containing two-edge collision. These assertions follow by substitution of the two local matrices on the preceding page. The exhaustive certificate enumerates the three two-edge orders (disjoint, crossed, nested), whether the left/right global extreme is present, and all legal contiguous marked intervals of either type. Its counts are

Local layouts (two or four endpoints)	248
Selected two-edge source equalities	241
Selected own-edge source equalities	489
Exterior L two-edge zero identities	241
Exterior L own-edge identities	489

All nilpotent masks are checked with exact rational arithmetic; the complete case inventory is saved with the certificate. This is a finite algebraic verification used together with the reduction lemma, not an extrapolation from a finite maximum field degree.

Let n_α count selected-boundary attachments inside interior L mark α. There is no inner-F nesting, and consequently

$$K_F(z) = K_G(z) - \sum_{\alpha:\, L} n_\alpha z_\alpha. \tag{3}$$

For the all-G word, add L singleton sources a, b at the two boundaries. The own-edge rule and two-edge zero identity give, when they are paired to distinct interior endpoints,

$$K_G(z, a, b) - a - b = K_G(z) + a\ell_0 + b\ell_L.$$

Here ℓ_0, ℓ_L are the interior L variables at the respective partners, or zero. For a direct exterior pair the right side is instead $K_G(z) + ab$. Insert $\chi = h - W$: each boundary applies $(1 - \partial_a)$ or $(1 - \partial_b)$ at zero to the exponential. The attached weight becomes $(1 - \lambda\ell_0)(1 - \lambda\ell_L)$, exactly (3). The direct exterior pair has weight $1 + \lambda$, whereas the selected direct pair has weight one.

Theorem 1 (Fixed-degree boundary identity)**.** *For every admissible expression B and finite linear combination thereof, with $s = \gamma^2 = \sqrt{q}$,*

$$\sigma_1(B) = [1 - s(1 + \lambda)]\tau(B) + s\tau(\chi B\chi), \qquad \chi = h - W. \tag{4}$$

Here τ denotes the limiting covariant central trace and σ_1 the limiting normalized single-selected expectation. In tracial moment notation the density is $\rho_1 = s\chi^2 + [1 - s(1 + \lambda)]I$.

The attached diagrams carry γ^2 and the direct diagrams carry one, which proves (4). This statement is on polynomial moments; it does not assert boundedness of all higher generators.

4 The charged tower and its exact pinching

If B contains only D marks and ordinary energy factors, both ℓ_0, ℓ_L vanish. Therefore

$$\sigma_1(B) = (1-s)\tau(B) + s\tau(hBh) = \tau(f(h)B), \qquad f(h) = 1 - s + sh^2. \tag{5}$$

This includes arbitrary fixed mixed words in all charged Wick descendants, not only the quadratic curvature. In particular the polynomial functional is invariant under the Hamiltonian derivation. Finite-time invariance for fixed insertions follows from the energy tail estimates below. This is not a temperature assignment.

Let $H_d^{(q)}$ be the monic q-Hermite polynomial with variance one, $P_d = [d]_q!$, and $m_d = q\partial_q \log P_d$. Use the finite-$N$ field

$$J_{d,N} = C_m^{-d/2}(dp - D)\mathcal{W}_d(H^d)|_m - \lambda_N[d(d-1) - 4m_d(q_N)]H_d^{(q_N)}(h_N), \quad q_N = e^{-4\lambda_N}. \tag{6}$$

For d=2 this is the preceding unnormalized curvature j_N. All normalization and regression coefficients are fixed before the double-scaling limit. The q-Hermite term commutes exactly with h_N.

For all-G diagrams, the full-trace local ratio is $2\sum_o F_o$, and both constant backgrounds are $-2k^2$. Thus the entire central marked-source exponent equals the earlier full-trace exponent, for arbitrary legal marks. Schur covariance equates the annealed fixed-Slater covariant expectation to normalized sector trace. This directly transfers the existing fixed-degree charged joint moments and spectral argument [4]; no extraction from a binomial charge average is used.

In particular, for fixed d,e the limiting two-energy measure

$$\nu_{de,N}(A,B) = \mathbb{E}\,\mathrm{tr}_m[J_{d,N}^\dagger P_{h_N}(B) J_{e,N} P_{h_N}(A)]$$

has the previously derived $L^2(\mu_q \otimes \mu_q)$ density. For normalized energy polynomials, the old bound reads, with $D = d + e$,

$$|M_{ab}^{de}| \le K_D(a+b+1)^D q^{\max(0,\min(a,b)-D)}\,\mathbf{1}_{|a-b|\le D}.$$

It is square summable for fixed d,e. The compact energy support is $[-2/\sqrt{1-q}, 2/\sqrt{1-q}]^2$; μ_q has no atoms, so the energy diagonal has zero mass. Central fixed higher moments give tightness and uniform integrability. Portmanteau then yields

$$\|\mathcal{P}_{h_N} J_{d,N}\|_{2,m} \longrightarrow 0, \qquad \mathcal{P}_h Y = \sum_E P_h(\{E\}) Y P_h(\{E\}). \tag{7}$$

Higher central moments and the Schatten contraction of pinching upgrade this to every fixed finite Schatten exponent. Complete degenerate blocks are retained. This takes finite-N exact pinching first and does not exchange a long-time averaging limit with double scaling.

5 Selected actual observables and the remaining time interface

The previously proved fixed-polynomial preparation estimate [5] gives, for any fixed preparation P_N with bounded $\mathbb{E}\,\|P_N O\|^2$,

$$\mathbb{E}\,\|X_N P_N O\|^2 \le C_P\,\|X_N\|_{4,m}^2\,.$$

Apply it to $X_N = \mathcal{P}_{h_N} J_{d,N}$. Since the new source theorem also bounds every fixed preparation norm, (7) implies

$$\mathbb{E}\,\big\|\,[(1-\mathcal{P}_{h_N})J_{d,N} - J_{d,N}]\,P_N O\big\|^2 \longrightarrow 0.$$

For single-selected covariant blocks, polarization, trace Holder and fixed higher moments similarly transfer any fixed finite number of these replacements. Fixed real-time energy phases are included by first approximating them in central trace norms and then applying the preparation bound. Indeed Gaussian energy moments obey $\|h_N^n\|_{r,m} \le K_r^n\sqrt{n!}$, making their Taylor tails summable at every fixed compact time interval after Holder.

Writing $a_{d,N} = (1-\mathcal{P}_{h_N})J_{d,N}$, the charged two-point matrix in the single-selected state is therefore

$$\lim_N \mathbb{E}\langle O|M_N a_{d,N}(t) a_{e,N} M_N^\dagger|O\rangle = \int f(E)e^{it(E-E')}\,d\nu_{de}(E,E'). \tag{8}$$

The input/column energy is E in this convention. Its positive matrix measure is $f(E)\nu_{de}$, and the D-only state is stationary. We do not replace every descendant's marginal by the tracial marginal: that further property was proved specifically for the curvature field.

The previous tagged-site proof [6] already supplies, sector by sector, for fixed r and a fixed number k of deficits,

$$\|\Delta_{T_1}\cdots\Delta_{T_k} Z_n\|_{r,m} \le K_{k,r}(n+1)^{2k} L_{k,r}^n\sqrt{n!}$$

in central charge ranges. It uses restricted covariance row/column bounds and keeps CAR signs until matrix contraction. Coupling OU projections, normalized covariance CP chains and their telescoping estimates also operate sectorwise. These estimates support pure/local representatives, including neutral Wick subblocks.

The ordinary charged coefficient needed by the actual full bilocal is

$$G_{n,N} = [p\mathcal{N} - D - \lambda_N\mathcal{N}(\mathcal{N}-1)]H^n.$$

It permits internal Gaussian pairs in its marked interval, outside the present theorem. Its existing full-Fock tail proof uses the Clifford Riesz estimate [2], whose trace is not the fixed-charge trace. Restriction to $m = N/2$ is not automatic. A central strong bridge and a degree-uniform actual tail, or an equivalent two-cutoff argument, are still needed to identify the full selected bilocal. Fixed-degree preparation constants cannot be summed over growing degrees without additional control. The previously completed full-trace bilocal four-point result is unchanged.

6 Evidence, reuse and continuation

The new pure module `slater_higher_wick_sources.py` accepts slot lengths, disjoint marks, Wick exclusion regions, chaos regions, scalar centers and trace/selected/χ modes. It is built on the already checked local CAR engine. Its coefficient exponential retains all nilpotent source masks. The fresh checker saves the complete finite local case inventory, 30 higher-layout density comparisons at two independent rational (γ, λ) points, and 18 comparisons against the previous H/W/J evaluator. Higher examples reach eight internal slots; they include two long D marks, neutral L subblocks, several simultaneous marks, and projected whole composites.

The rational points check algebraic identities; positivity and the physical state interpretation use $\gamma = e^{-\lambda}$. Finite examples alone do not prove all degrees: the finite local certificate is made exhaustive by the reduction lemma. The convergence and pinching claims also use the specified earlier estimates, not just this checker. Input hashes, Python version and exact arithmetic are saved with the new output. Successful prior scientific drivers and manifests are left unchanged.

The source extension is the new construction in this draft. Fixed-charge complex-SYK chord methods [1], the previous local source theorem, the charged spectral bound and the preparation-density estimate are reused with their original scopes. The unresolved ordinary central bridge is now the direct next step. Bulk spatial and mass generators, and finite-G dynamics, remain beyond this model-level result.

References

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[4] Alan Day and Chengheng, *Joint memory regression*, local project proof, 7 November 2026, Sections 5–7; `susskind-memory-joint-regression-proof.md`.

[5] Alan Day and Chengheng, *Fixed polynomial Slater preparations*, local project draft, 8 November 2026; `susskind-slater-polynomial-density`.

[6] Alan Day and Chengheng, *Bilocal fourth-tail diagram and operator drafts*, local project proofs, 7 November 2026; pure marked-block estimates and sectorwise CP bounds.